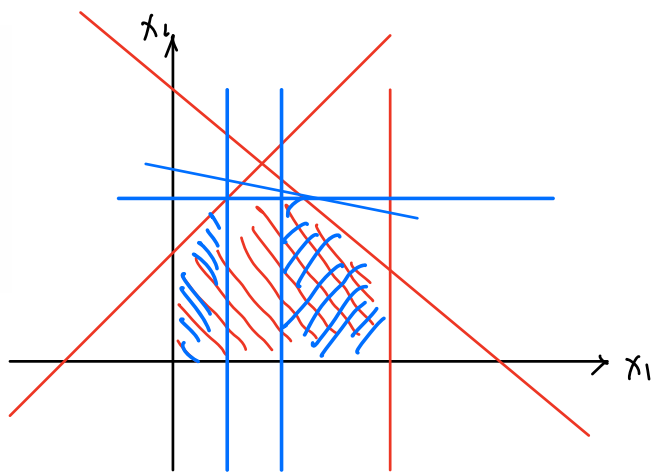


(1) 用分枝定界法求解下列整数规划问题

$$\begin{aligned} \min Z &= -x_1 - 5x_2 \\ \text{s.t. } &\begin{cases} x_1 - x_2 \geq -2 \\ 5x_1 + 6x_2 \leq 30 \\ x_1 \leq 4 \\ x_1, x_2 \geq 0 \text{ 且全为整数} \end{cases} \end{aligned}$$

$$\begin{cases} x_1 - x_2 = -2 \\ 5x_1 + 6x_2 = 30 \end{cases} \Rightarrow \begin{cases} x_1 = \frac{18}{11} \\ x_2 = \frac{40}{11} \end{cases}$$

$$Z^{(0)} = -\frac{18}{11} - 5 \times \frac{40}{11} = -\frac{218}{11}$$



$[\frac{18}{11}] = 1$. 分别增加约束 $x_1 \leq 1$ 和 $x_1 \geq 2$, 得到 (LP1) 和 (LP2)

$$\begin{aligned} \min Z &= -x_1 - 5x_2 \\ \text{(LP1) s.t. } &\begin{cases} x_1 - x_2 \geq -2 \\ 5x_1 + 6x_2 \leq 30 \\ 0 \leq x_1 \leq 1 \\ x_2 \geq 0 \end{cases} \end{aligned}$$

最优解 $(x_1, x_2)^T = (1, 3)^T$, $Z^{(1)} = -16$

$$\begin{aligned} \min Z &= -x_1 - 5x_2 \\ \text{(LP2) s.t. } &\begin{cases} x_1 - x_2 \geq -2 \\ 5x_1 + 6x_2 \leq 30 \\ 2 \leq x_1 \leq 4 \\ x_2 \geq 0 \end{cases} \end{aligned}$$

最优解 $(x_1, x_2)^T = (2, \frac{10}{3})^T$, $Z^{(2)} = -\frac{56}{3}$

$\because Z^{(2)} < Z^{(1)}$ $\therefore$ 对 (LP2) 进行分支

$\because [\frac{10}{3}] = 3$, 分别增加约束 $x_2 \leq 3$, $x_2 \geq 4$. 得到问题 (LP21) 和 (LP22)

$$\begin{aligned} \min Z &= -x_1 - 5x_2 \\ \text{(LP21) s.t. } &\begin{cases} x_1 - x_2 \geq -2 \\ 5x_1 + 6x_2 \leq 30 \\ 2 \leq x_1 \leq 4 \\ 0 \leq x_2 \leq 3 \end{cases} \end{aligned}$$

最优解 $(x_1, x_2)^T = (\frac{11}{2}, 3)^T$, $Z^{(21)} = -\frac{87}{2}$

$$\begin{aligned} \min Z &= -x_1 - 5x_2 \\ \text{(LP22) s.t. } &\begin{cases} x_1 - x_2 \geq -2 \\ 5x_1 + 6x_2 \leq 30 \\ 2 \leq x_1 \leq 4 \\ x_2 \geq 4 \end{cases} \end{aligned}$$

无可解

对 LP21 进行分支. $\because [\frac{11}{2}] = 2$. 分别增加约束 $x_1 \leq 2$, $x_1 \geq 3$. 得 (LP211) 和 (LP212)

$$\begin{aligned} \min Z &= -x_1 - 5x_2 \\ \text{(LP211) s.t. } &\begin{cases} x_1 - x_2 \geq -2 \\ 5x_1 + 6x_2 \leq 30 \\ x_1 = 2 \\ 0 \leq x_2 \leq 3 \end{cases} \end{aligned}$$

最优解 $(x_1, x_2)^T = (2, 3)^T$, $Z^{(211)} = -17$

$$\begin{aligned} \min Z &= -x_1 - 5x_2 \\ \text{(LP212) s.t. } &\begin{cases} x_1 - x_2 \geq -2 \\ 5x_1 + 6x_2 \leq 30 \\ 3 \leq x_1 \leq 4 \\ 0 \leq x_2 \leq 3 \end{cases} \end{aligned}$$

最优解 $(x_1, x_2)^T = (3, \frac{15}{2})^T$, $Z^{(212)} = -\frac{31}{2}$

$\because Z^{(212)} > Z^{(211)}$ $\therefore$ 原优化问题最优解为 $(x_1, x_2)^T = (2, 3)^T$, 最优值为 -17

课后练习题

(3) 已知效益矩阵如下，用匈牙利法求解该分配问题

$$\textcircled{1} \begin{pmatrix} 7 & 9 & 10 & 12 \\ 13 & 12 & 16 & 17 \\ 15 & 16 & 14 & 15 \\ 11 & 12 & 15 & 16 \end{pmatrix} \quad \textcircled{2} \begin{pmatrix} 3 & 8 & 2 & 10 & 3 \\ 8 & 7 & 2 & 9 & 7 \\ 6 & 4 & 2 & 7 & 5 \\ 8 & 4 & 2 & 3 & 5 \\ 9 & 10 & 6 & 9 & 10 \end{pmatrix}$$

$$\begin{aligned} \textcircled{1} \quad (C_{ij}^1) &= \begin{bmatrix} 0 & 2 & 3 & 5 \\ 1 & 0 & 4 & 5 \\ 1 & 2 & 0 & 1 \\ 0 & 1 & 4 & 5 \end{bmatrix} = \begin{bmatrix} 0 & 2 & 3 & 4 \\ 1 & 0 & 4 & 4 \\ 1 & 2 & 0 & 0 \\ 0 & 1 & 4 & 4 \end{bmatrix} = \begin{bmatrix} \textcircled{0} & 2 & 3 & 4 \\ 1 & \textcircled{0} & 4 & 4 \\ 1 & 2 & \textcircled{0} & \cancel{0} \\ \cancel{0} & 1 & 4 & 4 \end{bmatrix} \begin{matrix} \checkmark \\ \\ \\ \checkmark \end{matrix} \\ (C_{ij}^2) &= \begin{bmatrix} 0 & 1 & 2 & 3 \\ 2 & 0 & 4 & 4 \\ 2 & 2 & 0 & 0 \\ 0 & 0 & 3 & 3 \end{bmatrix} = \begin{bmatrix} \textcircled{0} & 1 & 2 & 3 \\ 2 & \textcircled{0} & 4 & 4 \\ 2 & 2 & \textcircled{0} & \cancel{0} \\ \cancel{0} & \cancel{0} & 3 & 3 \end{bmatrix} \begin{matrix} \checkmark \\ \checkmark \\ \\ \checkmark \end{matrix} \\ (C_{ij}^3) &= \begin{bmatrix} 0 & 1 & 0 & 1 \\ 2 & 0 & 2 & 2 \\ 4 & 4 & 0 & 0 \\ 0 & 0 & 1 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 1 & \textcircled{0} & 1 \\ 2 & \textcircled{0} & 2 & 2 \\ 4 & 4 & 0 & \textcircled{0} \\ \textcircled{0} & 0 & 1 & 1 \end{bmatrix} \end{aligned}$$

最大值 $z^* = 10 + 12 + 15 + 11 = 48$

课后练习题

(3) 已知效益矩阵如下，用匈牙利法求解该分配问题

$$\textcircled{1} \begin{pmatrix} 7 & 9 & 10 & 12 \\ 13 & 12 & 16 & 17 \\ 15 & 16 & 14 & 15 \\ 11 & 12 & 15 & 16 \end{pmatrix} \quad \textcircled{2} \begin{pmatrix} 3 & 8 & 2 & 10 & 3 \\ 8 & 7 & 2 & 9 & 7 \\ 6 & 4 & 2 & 7 & 5 \\ 8 & 4 & 2 & 3 & 5 \\ 9 & 10 & 6 & 9 & 10 \end{pmatrix}$$

$$\textcircled{2} \quad (C'_{ij}) = \begin{bmatrix} 0 & 4 & 0 & 8 & 1 \\ 5 & 3 & 0 & 7 & 5 \\ 3 & 0 & 0 & 5 & 3 \\ 5 & 0 & 0 & 1 & 3 \\ 2 & 2 & 0 & 3 & 4 \end{bmatrix} = \begin{bmatrix} \textcircled{0} & 4 & 0 & 7 & 1 \\ 5 & 3 & \textcircled{0} & 6 & 4 \\ 3 & \textcircled{0} & 0 & 4 & 2 \\ 5 & 0 & 0 & \textcircled{0} & 2 \\ 2 & 2 & 0 & 2 & 3 \end{bmatrix}$$

$$(C''_{ij}) = \begin{bmatrix} 0 & 4 & 2 & 7 & \textcircled{0} \\ 3 & 1 & \textcircled{0} & 4 & 2 \\ 3 & \textcircled{0} & 2 & 4 & 2 \\ 5 & 0 & 2 & \textcircled{0} & 2 \\ \textcircled{0} & 0 & 0 & 0 & 1 \end{bmatrix}$$

$$Z^* = 3 + 2 + 4 + 3 + 9 = 21$$

课后练习题

(2) 用割平面法求解下列整数规划问题

$$\begin{aligned} \max Z &= x_1 + x_2; \\ \text{s.t.} \quad &\begin{cases} 2x_1 + x_2 \leq 6; \\ 4x_1 + 5x_2 \leq 35; \\ x_1, x_2 \geq 0, \text{ 且为整数.} \end{cases} \end{aligned}$$

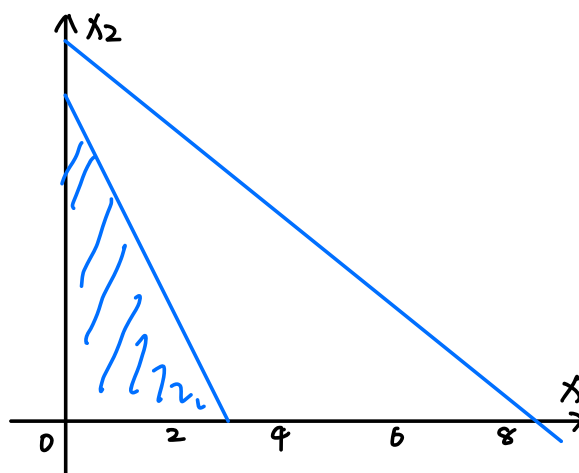
$$\max Z = x_1 + x_2$$

$$\Rightarrow \begin{cases} 2x_1 + x_2 + x_3 = 6 \\ 4x_1 + 5x_2 + x_4 = 35 \\ x_1, x_2, x_3, x_4 \geq 0, \text{ 且为整数} \end{cases}$$

	C		1	1	0	0
C_B	X_B	b	x_1	x_2	x_3	x_4
0	x_3	6	2	[1]	1	0
0	x_4	35	4	5	0	1
Z		0	1	1	0	0
1	x_2	6	2	1	1	0
0	x_4	5	-6	0	-5	1
Z		-6	-1	0	-1	0

$$\text{最优解 } (x_1, x_2)^T = (0, 6)^T$$

$$\text{此时 } Z^* = 6$$



课后练习题

(1) 某厂生产甲、乙、丙三种产品，需要消耗A、B两种原材料，已知每件产品对原材料的消耗、每件产品的成本及原材料的现有存量如右下表所示。要求制定生产计划，依次满足下列目标：

P_1 : 甲的产量要超过80;

P_2 : 丙的产量恰好达到100;

P_3 : 原材料A的现有量不得超过;

P_4 : 总成本限制在3*104元以下;

P_5 : 原材料B的现有量可以超过;

消耗 产品 \ 原料	A	B	单位 成本	需要量
甲	10	4	25	80以上
乙	6	8	25	
丙	8	12	30	100
现有量	1500	1600		

试建立此问题的目标规划模型。

设生产甲、乙、丙分别为 x_1, x_2, x_3 件，根据表格可得

$$\begin{cases} x_1 \geq 80 \\ x_3 = 100 \\ 10x_1 + 6x_2 + 8x_3 \leq 1500 \\ 4x_1 + 8x_2 + 12x_3 \leq 1600 \end{cases}$$

写出目标规划模型

$$\min z = P_1 d_1^- + P_2 (d_2^- + d_2^+) + P_3 d_3^+ + P_4 d_4^+ + P_5 d_5^-$$

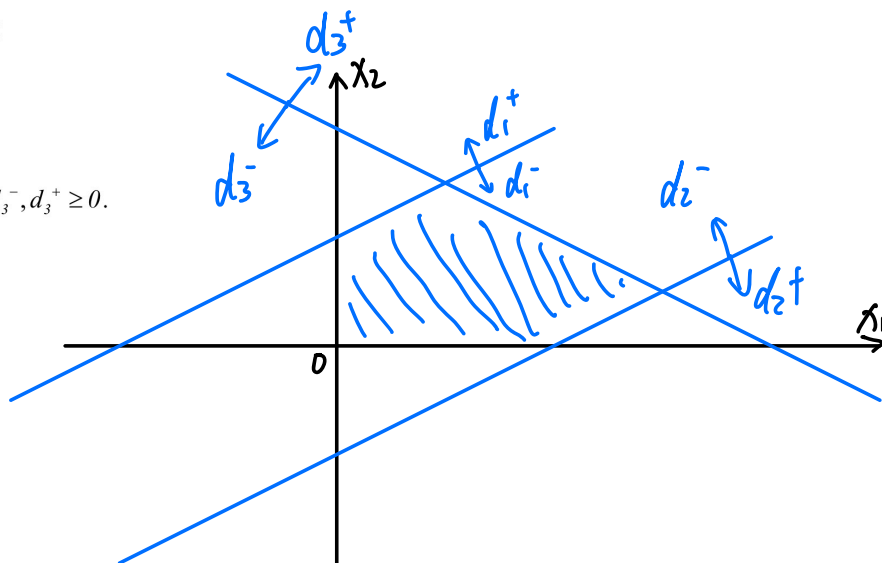
$$\text{s.t.} \begin{cases} x_1 + d_1^- - d_1^+ = 80 \\ x_3 + d_2^- - d_2^+ = 100 \\ 10x_1 + 6x_2 + 8x_3 + d_3^- - d_3^+ = 1500 \\ 25x_1 + 25x_2 + 30x_3 + d_4^- - d_4^+ = 3104 \\ 4x_1 + 8x_2 + 12x_3 + d_5^- - d_5^+ = 1600 \end{cases}$$

(2) 用图解法求解以下目标规划问题

① $\min Z = P_1 d_1^+ + P_2 d_2^+ + P_3 d_3^+;$

$$s.t. \begin{cases} -x_1 + 2x_2 + d_1^- - d_1^+ = 4; \\ x_1 - 2x_2 + d_2^- - d_2^+ = 4; \\ x_1 + 2x_2 + d_3^- - d_3^+ = 8; \\ x_1, x_2, d_1^-, d_1^+, d_2^-, d_2^+, d_3^-, d_3^+ \geq 0. \end{cases}$$

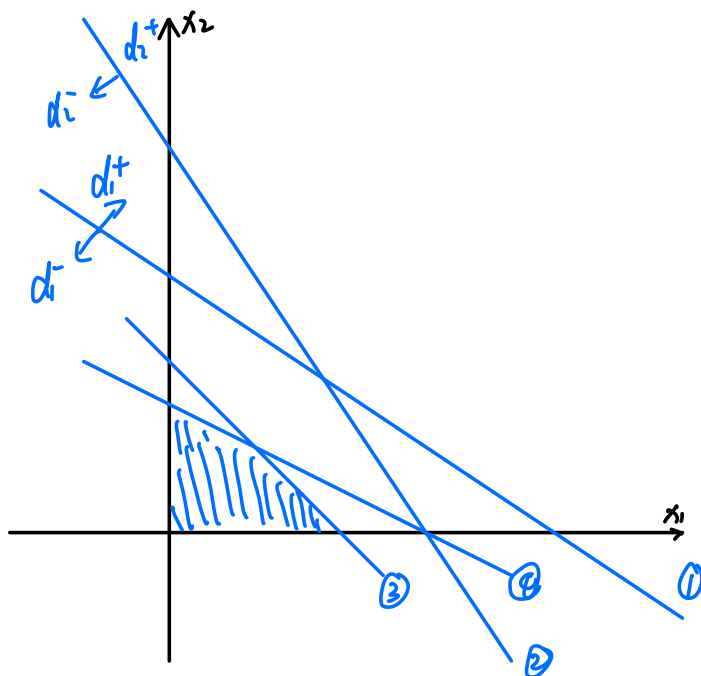
图中阴影范围即
为满意解



② $\min Z = P_1 (d_1^+ + d_1^-) + P_2 (d_2^+ + d_2^-);$

$$s.t. \begin{cases} 2x_1 + 3x_2 + d_1^- - d_1^+ = 18; \\ 3x_1 + 2x_2 + d_2^- - d_2^+ = 18; \\ x_1 + x_2 \leq 4; \\ x_1 + 2x_2 \leq 6; \\ x_1, x_2, d_1^-, d_1^+, d_2^-, d_2^+ \geq 0. \end{cases}$$

先考虑第一个目标, 可知
问题求阴影范围离直
线①最近的点.
即 $(x_1, x_2)^T = (2, 2)^T$



(3) 用单纯形法求解下列目标规划问题

① $\min Z = P_1 d_2^+ + P_1 d_2^- + P_2 d_1^-;$

$$s.t. \begin{cases} x_1 + 2x_2 + d_1^- - d_1^+ = 10; \\ 10x_1 + 12x_2 + d_2^- - d_2^+ = 62.4; \\ 2x_1 + x_2 \leq 8; \\ x_1, x_2, d_1^-, d_1^+, d_2^-, d_2^+ \geq 0. \end{cases}$$

② $\min Z = P_1(d_1^+ + d_2^+) + P_2 d_3^-;$

$$s.t. \begin{cases} x_1 + x_2 + d_1^- - d_1^+ = 1; \\ 2x_1 + 2x_2 + d_2^- - d_2^+ = 4; \\ 6x_1 - 4x_2 + d_3^- - d_3^+ = 50; \\ x_1, x_2, d_i^-, d_i^+ (i=1,2,3) \geq 0. \end{cases}$$

① 列出单纯形表求解

	C		0	0	0	P ₂	0	P ₁	P ₁
C _B	X _B	b	x ₁	x ₂	x ₃	d ₁ ⁻	d ₁ ⁺	d ₂ ⁻	d ₂ ⁺
P ₂	d ₁ ⁻	10	1	[2]	0	1	-1	0	0
P ₁	d ₂ ⁻	62.4	10	12	0	0	0	1	-1
0	x ₃	8	2	1	1	0	0	0	0
σ _{kj}	P ₁	-62.4	-10	-12	0	0	0	0	2
	P ₂	-10	-1	-2	0	0	1	0	0
0	x ₂	5	1/2	1	0	1/2	-1/2	0	0
P ₁	d ₁ ⁻	2.4	4	0	0	-6	[6]	1	-1
0	x ₃	3	3/2	0	1	-1/2	1/2	0	0
σ _{kj}	P ₁	-2.4	-4	0	0	6	-6	0	2
	P ₂	0	0	0	0	1	0	0	0
0	x ₂	26/5	5/6	1	0	0	0	1/2	-1/2
0	d ₁ ⁺	2/5	2/3	0	0	-1	1	1/6	-1/6
0	x ₃	18/5	7/6	0	1	0	0	-1/2	1/2
σ _{kj}	P ₁	0	0	0	0	0	0	1	1
	P ₂	0	0	0	0	1	0	0	0

∵ a₁ = a₂ = 0, 得到满意解

此时 $(x_1, x_2, x_3)^T = (0, 26/5, 18/5)^T, d_1^+ = 2/5, Z^* = 0$

(3) 用单纯形法求解下列目标规划问题

① $\min Z = P_1 d_2^+ + P_1 d_2^- + P_2 d_1^-;$

$$s.t. \begin{cases} x_1 + 2x_2 + d_1^- - d_1^+ = 10; \\ 10x_1 + 12x_2 + d_2^- - d_2^+ = 62.4; \\ 2x_1 + x_2 \leq 8; \\ x_1, x_2, d_1^-, d_1^+, d_2^-, d_2^+ \geq 0. \end{cases}$$

② $\min Z = P_1(d_1^+ + d_2^+) + P_2 d_3^-;$

$$s.t. \begin{cases} x_1 + x_2 + d_1^- - d_1^+ = 1; \\ 2x_1 + 2x_2 + d_2^- - d_2^+ = 4; \\ 6x_1 - 4x_2 + d_3^- - d_3^+ = 50; \\ x_1, x_2, d_i^-, d_i^+ (i=1,2,3) \geq 0. \end{cases}$$

② 列出单纯形表求解

	C		0	0	0	P_1	0	P_1	P_2	0
C_B	X_B	b	x_1	x_2	d_1^-	d_1^+	d_2^-	d_2^+	d_3^-	d_3^+
0	d_1^-	1	[1]	1	1	-1	0	0	0	0
0	d_2^-	4	2	2	0	0	1	-1	0	0
P_2	d_3^-	50	6	-4	0	0	0	0	1	-1
σ_j	P_1	0	0	0	0	1	0	1	0	0
	P_2	-50	-6	4	0	0	0	0	0	1
0	x_1	1	1	1	1	-1	0	0	0	0
0	d_2^-	2	0	0	-2	[2]	1	-1	0	0
P_2	d_3^-	44	0	-10	-6	6	0	0	1	-1
σ_j	P_1	0	0	0	0	1	0	1	0	0
	P_2	-44	0	10	6	-6	0	0	0	1

∵ P_2 全部负检验数上均有正检验数, 且 a_1, a_2 均大于 0,

∴ 已求得满意解

此时 $(x_1, x_2)^T = (1, 0)^T, d_2^- = 2, d_3^- = 44, Z^* = 44P_2$

$$\max Z = 4x_1 + 9x_2 + 2x_3^2$$

$$\text{s.t. } \begin{cases} x_1 + x_2 + x_3 = 10 \\ x_i \geq 0 \end{cases}$$

$$x_1 = s_1, s_2 = s_1 + x_2, s_3 = s_2 + x_3$$

$$f_1(s_1) = \max_{x_1=s_1} \{4x_1\} = 4s_1$$

$$\begin{aligned} f_2(s_2) &= \max \{9x_2 + 4s_1\} \\ &= \max \{9x_2 + 4s_2 - 4x_2\} \\ &= \max \{5x_2 + 4s_2\} = 9s_2 \end{aligned}$$

$$\begin{aligned} f_3(s_3) &= \max_{0 \leq x_3 \leq s_3} \{2x_3^2 + 9s_3 - 9x_3\} \\ &= \max \{9s_3, 2s_3^2\} \end{aligned}$$

$$\max f_3(s_3) = 200$$

$$\text{at } s_3 = 10, s_1 = s_2 = 0$$

$$\therefore x = (0, 0, 10)^T$$

$$x_1 = \frac{s_1}{3}, s_2 = s_1 + 4x_2, s_3 = s_2 + 5x_3$$

$$f_1(s_1) = \max_{x_1=\frac{s_1}{3}} \{4x_1\} = \frac{4}{3}s_1$$

$$\begin{aligned} f_2(s_2) &= \max_{0 \leq x_2 \leq \frac{s_2}{4}} \{5x_2 + f_1(s_1)\} = \max \{5x_2 + \frac{4}{3}s_1\} \\ &= \max \{5x_2 + \frac{4}{3}(s_2 - 4x_2)\} \\ &= \max_{0 \leq x_2 \leq \frac{s_2}{4}} \{5x_2 + \frac{4}{3}s_2 - \frac{16}{3}x_2\} \\ &= \frac{4}{3}s_2 \end{aligned}$$

$$f_3(s_3) = \max_{0 \leq x_3 \leq \frac{s_3}{5}} \{6x_3 + \frac{4}{3}(s_3 - 5x_3)\} = \frac{4}{3}s_3$$

$$0 \leq s_3 \leq 10$$

$$\therefore s_3 = 10, s_2 = s_1 = 0$$

$$(x_1, x_2, x_3)^T = (\frac{10}{3}, 0, 0)^T$$

$$\begin{aligned} \textcircled{2} \max Z &= 3x_1^3 - 4x_1 + 2x_2^2 - 5x_2 + 2x_3 \\ \text{s.t. } &\begin{cases} 4x_1 + 2x_2 + 3x_3 \leq 18 \\ x_i \geq 0 \end{cases} \end{aligned}$$

$$f_1(s_1) = \max_{x_1=\frac{s_1}{4}} \{3x_1^3 - 4x_1\} = \frac{3s_1^3}{64} - s_1$$

$$f_2(s_2) = \max \{2x_2^2 - 5x_2 + \frac{3}{64}(s_1 - 2x_2)^3 - (s_1 - x_2)\}$$

$$f_4(s_4) = 0$$

$$f_3(s_3) = \max_{x_3=s_3} \{2x_3^2 + f_4(s_4)\} = 2s_3^2$$

$$\begin{aligned} f_2(s_2) &= \max_{0 \leq x_2 \leq s_2} \{9x_2 + f_3(s_3)\} = \max \{9x_2 + 2s_3^2\} \\ s_3 &= s_2 - x_2 \end{aligned}$$

$$\begin{aligned} f_2(s_2) &= \max \{9x_2 + 2s_2^2 - 4s_2x_2 + 2x_2^2\} \\ &= \max \{2s_2^2, 9s_2\} \end{aligned}$$

$$f_1(s_1) = \max_{0 \leq x_1 \leq s_1} \{4x_1 + f_2(s_2)\} \quad s_2 = s_1 - x_1$$

$$\begin{aligned} \textcircled{1} f_1(s_1) &= \max \{4x_1 + 2s_1^2 - 4s_1x_1 + 2x_1^2\} \\ &= \max \{9s_1, 2s_1^2\} \quad s_1 \in [0, 10] \end{aligned}$$

$$f_1(s_1) = 200, s_1 = 10$$

$$\begin{aligned} \textcircled{2} f_1(s_1) &= \max_{0 \leq x_1 \leq s_1} \{4x_1 + 9(s_1 - x_1)\} \\ &= \max \{9s_1 - 5x_1\} = 9s_1 \end{aligned}$$

$$\therefore f_1(s_1) = 200, s_1 = x_1 = 10, x_2 = x_3 = 0$$